# INNOPOLIS UNIVERSITY

## SNE MASTER TRACK

## 2025

## COURSE: [F25] Networks Engineering

### NE Lab 6 Report

**On**

### BGP

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**Part A: Basic Router Configuration**

**1. Configure hostname according to its AS number**

**Answer:**

**AS Numbers**

In the preparation phase of this lab, it was very important to select the Autonomous System (AS) number correctly. As per the lab instructions, the three routers have to be configured in three different ASs, where the AS number format will be AS650[XX]1, AS650[XX]2 and AS650[XX]3. Here the [XX] part is determined according to the Student ID. My Student ID is 24, so accordingly I used the following AS numbers for the routers:

Router-1: AS650241, Router-2: AS650242 and Router-3: AS650243. These AS numbers follow the naming convention specified in the lab completely. As a result, each router can function properly as a separate Autonomous System and the necessary conditions for eBGP (external BGP) configuration are met. Due to the use of the correct AS number, no conflict or error was observed in BGP neighbor establishment.

**Loopback Subnets (/28) + Loopback IP (/32)**

As per the lab instructions, it is mandatory to configure one loopback interface on each router and later used for BGP router-ID and reachability test. As a result, a separate /32 loopback IP has been assigned for each Autonomous System.

In my setup, 10.24.1.1/32 loopback IP has been configured for Router-1 (AS650241), 10.24.2.1/32 for Router-2 (AS650242) and 10.24.3.1/32 for Router-3 (AS650243). Each loopback IP is part of the /28 subnet. Loopback interfaces are not only used as router-ID, but also play an important role in verifying end-to-end connectivity.

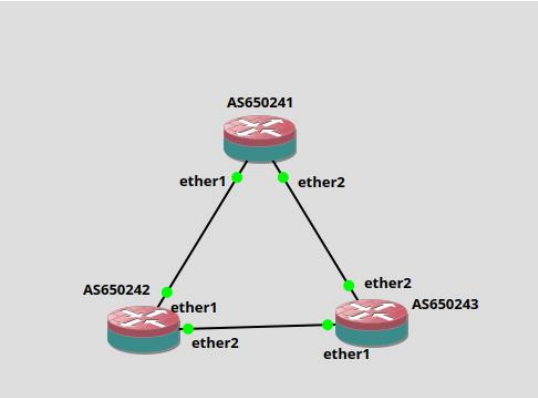
**Transit Networks (/30)**

**My setup (Student ID 24):**

| Link | Network | IPs Used |
|---|---|---|
| AS241-242 | 172.16.24.0/30 | .1 / .2 |
| AS242-243 | 172.16.24.4/30 | .5 / .6 |

| Link | Network | IPs Used |
|---|---|---|
| AS243-241 | 172.16.24.8/30 | .9 / .10 |

**Network Topology:**




**Routers Configuration**

The GNS3 environment is configured with three routers, where each router represents a separate Autonomous System (AS). For this purpose, I used three MikroTik CHR routers, which are fully suitable for the lab and support all the features of BGP configuration.

**Configure hostname according to AS number**

The hostname of each router is set according to the corresponding AS number so that the network topology is easy to understand and there is no problem during verification. The names of the routers are set to AS650241, AS650242 and AS650243 respectively using the */system identity set name=* command of RouterOS.

**Hostname:**

**AS650241**

```
[admin@MikroTik] > /system identity set name=AS650241
```

**AS650242**

```
[admin@MikroTik] > /system identity set name=AS650242
```

**AS650243**

```
[admin@MikroTik] > /system identity set name=AS650243
```

**2. Configure loopback interface with the assigned /32 address**

**Answer:**

**Configure loopback interface with /32**

**AS650241**

***Adding name=loopback***

```
[admin@AS650241] > /interface bridge add name=loopback
```

***Adding address=10.24.1.1/32 and setting interface=loopback***

```
[admin@AS650241] > /ip address add address=10.24.1.1/32 interface=loopback
```

**AS650242**

***Adding name=loopback***

```
[admin@AS650242] > /interface bridge add name=loopback
```

***Adding address=10.24.2.1/32 and setting interface=loopback***

```
[admin@AS650242] > /ip address add address=10.24.2.1/32 interface=loopback
```

**AS650243**

***Adding name=loopback***

```
[admin@AS650243] > /interface bridge add name=loopback
```

***Adding address=10.24.3.1/32 and setting interface=loopback***

```
[admin@AS650243] > /ip address add address=10.24.3.1/32 interface=loopback
```

Here, loopback interfaces have been configured on all three routers and assigned a specific /32 IP address. Since there is no separate loopback interface directly in MikroTik RouterOS, loopback has been created using a bridge interface. A bridge interface named loopback has been created on Router-1 (AS650241) and the *10.24.1.1/32* IP address has been added there. *10.24.2.1/32* has been configured on Router-2 (AS650242) and *10.24.3.1/32* loopback IP has been configured on Router-3 (AS650243).

**3. Configure physical interfaces with appropriate IP addresses from the /30 transit networks**

**Answer:**

**Configure physical interfaces with /30 IPs**

Here, the physical (Ethernet) interfaces of the routers are configured to use a /30 transit network for communication between each system. The appropriate IP address is assigned to the specific Ethernet interface on each router using the /ip address add command. This command adds an IP address and subnet mask to the corresponding interface so that the routers can communicate directly with each other.

**AS650241**

The ether1 interface on Router-1 (AS650241) is connected to Router-2 (AS650242) and is configured with the *172.16.24.1/30* IP address, which is part of the *172.16.24.0/30* subnet.

```
[admin@AS650241] > /ip address add address=172.16.24.1/30 interface=ether1
```

Also, the ether2 interface on this router is connected to Router-3 (AS650243) and the IP address *172.16.24.10/30* is set there, which is part of the *172.16.24.8/30* subnet.

```
[admin@AS650241] > /ip address add address=172.16.24.10/30 interface=ether2
```

**AS650242**

In the case of AS650242, the ether1 interface uses the IP *172.16.24.2/30*, which creates a point-to-point link with Router-1.

```
[admin@AS650242] > /ip address add address=172.16.24.2/30 interface=ether1
```

The ether2 interface has the 172.16.24.5/30 IP configured which connects to Router-3 and uses the *172.16.24.4/30* subnet.

```
[admin@AS650242] > /ip address add address=172.16.24.5/30 interface=ether2
```

**AS650243**

The ether1 interface on AS650243 is assigned the IP *172.16.24.6/30* which is connected to Router-2.

```
[admin@AS650243] > /ip address add address=172.16.24.6/30 interface=ether1
```

A point-to-point link has been created with Router-1 using the *172.16.24.9/30* IP on the ether2 interface.

```
[admin@AS650243] > /ip address add address=172.16.24.9/30 interface=ether2
```

This way each physical interface is configured to correspond to the correct subnet and the correct neighboring router.

**4. Enable routing and ensure interfaces are active**

**Answer:**

**Enable routing & ensure interfaces active**

The main purpose of this part was to check whether routing is enabled on each router and whether all the required interfaces are up. IP routing is enabled by default in MikroTik RouterOS, so no separate command is required to enable routing. When an IP address is assigned to an interface, that interface automatically goes into the enabled (running) state.

According to the output below, running the /ip address print command on all three routers shows that the /32 IP on the loopback interface and the /30 transit network IP on the physical interfaces are successfully configured. This means that the loopback, ether1 and ether2 interfaces are up. Checking the interface status using the /interface print command shows that all the required interfaces are not disabled and are ready for routing. This configuration clearly shows that routing is enabled for proper operation on each router and no interfaces are in shutdown state.

**Verification:**

**AS650241**

```
[admin@AS650241] > /ip address print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK      INTERFACE
0 10.24.1.1/32      10.24.1.1    loopback
1 172.16.24.1/30    172.16.24.0  ether1
2 172.16.24.10/30   172.16.24.8  ether2
[admin@AS650241] >
```

**AS650242**

```
[admin@AS650242] > /ip address print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS           NETWORK      INTERFACE
0 10.24.2.1/32      10.24.2.1    loopback
1 172.16.24.2/30    172.16.24.0  ether1
2 172.16.24.5/30    172.16.24.4  ether2
[admin@AS650242] >
```

**AS650243**

```
[admin@AS650243] > /ip address print
Columns: ADDRESS, NETWORK, INTERFACE
# ADDRESS          NETWORK      INTERFACE
0 10.24.3.1/32     10.24.3.1    loopback
1 172.16.24.6/30   172.16.24.4  ether1
2 172.16.24.9/30   172.16.24.8  ether2
[admin@AS650243] >
```

**Part B - BGP Configuration**

**1. Enable BGP routing with the correct AS number for each router**

**Answer:**

BGP routing has been enabled on each MikroTik router and the correct AS number has been used for each router as per the lab instructions. The routing bgp template is used to enable BGP in RouterOS v7.

**Config:**

**As=6502241**

**Enabling BGP with correct AS**

A BGP template named TPL-AS650241 has been created on the AS650241 router, where the correct AS number is specified as as=650241 and the loopback IP is used as router-id=10.24.1.1. Also, the routing table is selected as main and connected and static routes are allowed to be advertised via BGP by setting *output.redistribute=connected,static*. Using *output.network=yes* causes the valid networks in the routing table to be advertised via BGP.

```
[admin@AS650241] > /routing bgp template add \
\... name=TPL-AS650241 \
\... as=650241 \
\... router-id=10.24.1.1 \
\... routing-table=main \
\... output.redistribute=connected,static \
\... output.network=yes
```

**as=650242**

The TPL-AS650242 template has been created on the AS650242 router, where as=650242 and *router-id=10.24.2.*1 have been used. The TPL-AS650243 template has been created on the AS650243 router, following the same structure, and as=650243 and *router-id=10.24.3.1* have been configured.

```
[admin@AS650242] > /routing bgp template add \
\... name=TPL-AS650242 \
\... as=650242 \
\... router-id=10.24.2.1 \
\... routing-table=main \
\... output.redistribute=connected,static \
\... output.network=yes
```

**as=650243**

To enable BGP on router AS650243, a BGP template has been configured, where a template named TPL-AS650243 has been created. The correct Autonomous System number for this router has been determined using as=650243.

In this configuration, *router-id=10.24.3.*1 has been set, which is the /32 IP address of the router's loopback interface. The reason for using the loopback IP is that it is permanent and does not change even if the interface is down, which is a best practice for BGP. Using *routing-table=main* has ensured that only the routes in the main routing table are used in the BGP process. Using the *output.redistribute=connected,static* option, the router's connected route (loopback /32) and static route (later added /28 blackhole route) have been allowed to be advertised by BGP. Also, since *output.network=yes* is enabled, it has been possible to automatically send valid networks in the routing table as BGP updates.

```
[admin@AS650243] > /routing bgp template add \
\... name=TPL-AS650243 \
\... as=650243 \
\... router-id=10.24.3.1 \
\... routing-table=main \
\... output.redistribute=connected,static \
\... output.network=yes
```

**2. Set router ID to match the loopback address**

**Answer:**

In this step, we were asked to configure the BGP Router-ID for each router so that it exactly matches the /32 IP address of the corresponding loopback interface. As per the lab requirement,

the Router-ID must be a permanent and unique IP address, because the loopback interface never goes down and the identity of the router always remains the same.

In this lab, the Router-ID for the AS650241 router is *10.24.1.1*, which is the IP address of its loopback interface.

```
[admin@AS650241] > /routing bgp template add \
\... name=TPL-AS650241 \
\... as=650241 \
\... router-id=10.24.1.1 \
\... routing-table=main \
\... output.redistribute=connected,static \
\... output.network=yes
```

The Router-ID is set to *10.24.2.1* on the AS650242 router and *10.24.3.1* on the AS650243 router.

```
[admin@AS650242] > /routing bgp template add \
\... name=TPL-AS650242 \
\... as=650242 \
\... router-id=10.24.2.1 \
\... routing-table=main \
\... output.redistribute=connected,static \
\... output.network=yes
```

```
[admin@AS650243] > /routing bgp template add \
\... name=TPL-AS650243 \
\... as=650243 \
\... router-id=10.24.3.1 \
\... routing-table=main \
\... output.redistribute=connected,static \
\... output.network=yes
```

These three IP addresses are /32 loopback addresses, which makes each router's Router-ID unique and permanent.

This configuration was done using the router-id parameter while creating the BGP template in RouterOS v7. As a result, BGP neighbor establishment, route advertisement, and BGP path selection all worked correctly and there were no Router-ID conflicts.

### 3. Configure eBGP neighbors pointing to the other routers' interface Ips

**Answer:**

In this step, we have to configure eBGP (inter-AS) neighbors for three routers. The neighbor must be the /30 IP address of the physical interface of the other router, not the loopback. In addition, the correct remote AS number is specified for each neighbor.

**as=S650241**

Two eBGP neighbors are configured on this router, one with AS650242 and the other with AS650243. The first command uses the TPL-AS650241 template, which indicates the BGP configuration of this router. Here, *local.address=172.16.24.1* is given because it is the physical interface IP of the AS650241 router that is connected to AS650242. *remote.address=172.16.24.2* is the interface IP of the AS650242 router. The AS number of the neighboring router is specified with remote.as=650242 and *local.role=ebgp* is set to ensure that this is an inter-AS BGP session.

```
[admin@AS650241] > /routing bgp connection add \
\... name=EBGP-AS650242 \
\... template=TPL-AS650241 \
\... local.address=172.16.24.1 \
\... remote.address=172.16.24.2 \
\... remote.as=650242 \
\... local.role=ebgp
```

In the second command, an eBGP neighbor has been created with AS650243 from the same router. Here, *local.address=172.16.24.10* has been used which is the other physical interface IP of AS650241 and *remote.address=172.16.24.9* is the interface IP of AS650243 router. The correct AS has been specified with remote.as=650243. As a result, AS650241 router has been able to establish eBGP peering correctly with the routers on both sides.

```
[admin@AS650241] > /routing bgp connection add \
\... name=EBGP-AS650243 \
\... template=TPL-AS650241 \
\... local.address=172.16.24.10 \
\... remote.address=172.16.24.9 \
\... remote.as=650243 \
\... local.role=ebgp
```

**as=650242**

In this router, the first eBGP connection is peering from AS650242 to AS650241. Here, *local.address=172.16.24.2* is used which is the physical interface IP of the AS650242 router and *remote.address=172.16.24.1* is the interface IP of the AS650241 router. The AS number of the neighboring router is specified with remote.as=650241 and *local.role=ebgp* is set.

```
[admin@AS650242] > /routing bgp connection add \
\... name=EBGP-AS650241 \
\... template=TPL-AS650242 \
\... local.address=172.16.24.2 \
\... remote.address=172.16.24.1 \
\... remote.as=650241 \
\... local.role=ebgp
```

In the second connection, an eBGP session is created from AS650242 to AS650243. Here, *local.address=172.16.24.5* and *remote.address=172.16.24.6* are used, which are the IP pairs of the

/30 transit network between the AS650242 and AS650243 routers. In this way, the AS650242 router has correctly established eBGP neighbors with its two neighboring ASs.

```
[admin@AS650242] > /routing bgp connection add \
\... name=EBGP-AS650243 \
\... template=TPL-AS650242 \
\... local.address=172.16.24.5 \
\... remote.address=172.16.24.6 \
\... remote.as=650243 \
\... local.role=ebgp
```

**as=650243**

The eBGP neighbor is configured on the AS650243 router with the same policy. In the first command, an eBGP session is created from AS650243 to AS650241. Here, *local.address=172.16.24.9* is used which is the physical interface IP of the AS650243 router and *remote.address=172.16.24.10* is the interface IP of the AS650241 router. The correct AS is specified with remote.as=650241 and *local.role=ebgp* is set.

```
[admin@AS650243] > /routing bgp connection add \
\... name=EBGP-AS650241 \
\... template=TPL-AS650243 \
\... local.address=172.16.24.9 \
\... remote.address=172.16.24.10 \
\... remote.as=650241 \
\... local.role=ebgp
```

Then the command creates an eBGP neighbor from AS650243 to AS650242. Here *local.address=172.16.24.6* and *remote.address=172.16.24.5* are used, which are the IP of the /30 transit link between AS650243 and AS650242 routers. Through this, AS650243 router has completed inter-AS BGP peering with its two neighboring routers.

```
[admin@AS650243] > /routing bgp connection add \
\... name=EBGP-AS650242 \
\... template=TPL-AS650243 \
\... local.address=172.16.24.6 \
\... remote.address=172.16.24.5 \
\... remote.as=650242 \
\... local.role=ebgp
```

**4. Activate BGP for IPv4 unicast on all neighbor relationships**

**Answer:**

This step ensures that IPv4 unicast address-family is enabled for each BGP neighbor relationship. In RouterOS v7, IPv4 unicast address-family is enabled by default, so there is no need to issue a separate address-family ipv4 unicast command.

In this lab, IPv4 unicast activation is not done in a separate step, but rather in the previous step, Task 2 Question 1. When creating the BGP template, *output.network=yes* and *output.redistribute=connected,static* were configured. Through this, all IPv4 networks in the routing table-especially connected routes (loopback /32) and static routes (loopback subnet /28)-are automatically advertised as IPv4 unicast by BGP.

**AS650241**

```
[admin@AS650241] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650242"
     remote.address=172.16.24.2 .as=650242
     local.address=172.16.24.1 .role=ebgp
     routing-table=main router-id=10.24.1.1 templates=TPL-AS650241 as=650241
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650243"
     remote.address=172.16.24.9 .as=650243
     local.address=172.16.24.10 .role=ebgp
     routing-table=main router-id=10.24.1.1 templates=TPL-AS650241 as=650241
     output.redistribute=connected,static .network=yes
```

```
[admin@AS650242] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.1/32 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10
```

**AS650242**

```
[admin@AS650242] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0    name="EBGP-AS650241"
      remote.address=172.16.24.1 .as=650241
      local.address=172.16.24.2 .role=ebgp
      routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
      output.redistribute=connected,static .network=yes

 1    name="EBGP-AS650243"
      remote.address=172.16.24.6 .as=650243
      local.address=172.16.24.5 .role=ebgp
      routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
      output.redistribute=connected,static .network=yes
```

```
[admin@AS650243] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10
```

**AS650243**

```
[admin@AS650243] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0    name="EBGP-AS650241"
      remote.address=172.16.24.10 .as=650241
      local.address=172.16.24.9 .role=ebgp
      routing-table=main router-id=10.24.3.1 templates=TPL-AS650243 as=650243
      output.redistribute=connected,static .network=yes

 1    name="EBGP-AS650242"
      remote.address=172.16.24.5 .as=650242
      local.address=172.16.24.6 .role=ebgp
      routing-table=main router-id=10.24.3.1 templates=TPL-AS650243 as=650243
      output.redistribute=connected,static .network=yes
```

```
[admin@AS650243] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10
```

Since IPv4 unicast is enabled by default in RouterOS v7 and the required advertise settings are included in the configuration in the previous question, there is no need to run any additional commands for this question. All BGP neighbor relationships are already capable of IPv4 unicast route exchange.

**5. Advertise the /28 loopback subnet and the /32 loopback address**

**Answer:**

Each router is advertising its loopback IP (/32) and loopback subnet (/28) to other ASs via BGP. In RouterOS v7, a separate network command is not used; rather, BGP advertisement is accomplished by redistributing routes in the routing table. A loopback interface has been created on all three routers and a /32 IP address has been assigned to it. In RouterOS, a bridge interface is typically used as a loopback.

**AS650241**

A logical loopback interface has been created.

```
[admin@AS650241] > /interface bridge add name=loopback
```

A specific /32 loopback IP has been assigned to the AS650241 router.

```
[admin@AS650241] > /ip address add address=10.24.1.1/32 interface=loopback
```

**AS650242**

```
[admin@AS650242] > /interface bridge add name=loopback
```

In this router, 10.24.2.1/32 loopback IP has been added to the routing table as a connected route.

```
[admin@AS650242] > /ip address add address=10.24.2.1/32 interface=loopback
```

**AS650243**

```
[admin@AS650243] > /interface bridge add name=loopback
```

On this router, 10.24.3.1/32 loopback IP has been added to the routing table as a connected route.

```
[admin@AS650243] > /ip address add address=10.24.3.1/32 interface=loopback
```

A static blackhole route is added to each of the three routers to advertise the /28 subnet. The purpose of this is to keep the subnet present in the routing table so that BGP can advertise it. Since the /28 subnet is not actually used directly on any interface, it is added as a static blackhole route.

**AS650241**

*dst-address=10.24.1.0/28* is the subnet to advertise and the blackhole drops traffic but the route remains in the routing table. As a result, the /28 subnet is added as a static route.

```
[admin@AS650241] > /ip route add dst-address=10.24.1.0/28 blackhole comment="Lab6 BGP advertise /28"
```

**AS650242**

```
[admin@AS650242] > /ip route add dst-address=10.24.2.0/28 blackhole comment="Lab6 BGP advertise /28"
```

**AS650243**

```
[admin@AS650243] > /ip route add dst-address=10.24.3.0/28 blackhole comment="Lab6 BGP advertise /28"
```

With this setting, all the static routes (*blackhole /28 subnet*) in the routing table are advertised in BGP. As a result, the /28 loopback subnet is successfully advertised to all neighbors.

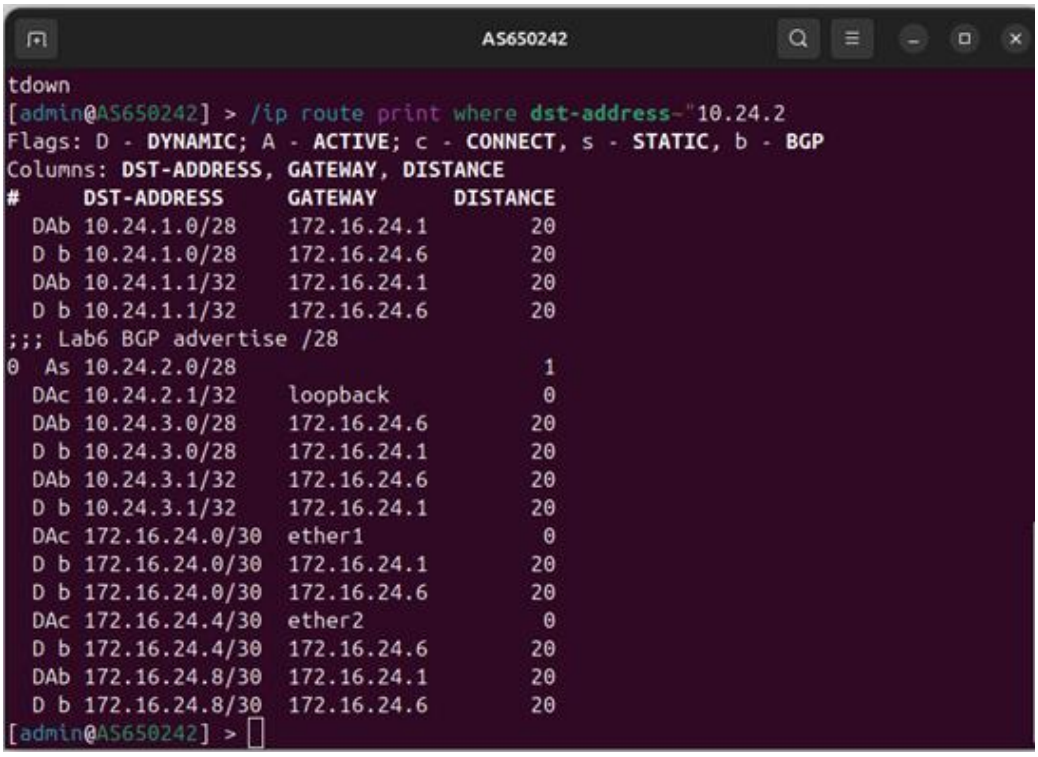
**VERIFY**

After running the following command on all the routers, the output shows that /32 is DAC (connected) route and /28 is S (static blackhole) route. Also, the *Db / DAb* flag is visible in the BGP learned routes. This proves that both /32 and /28 are present in the routing table and are being advertised through BGP.

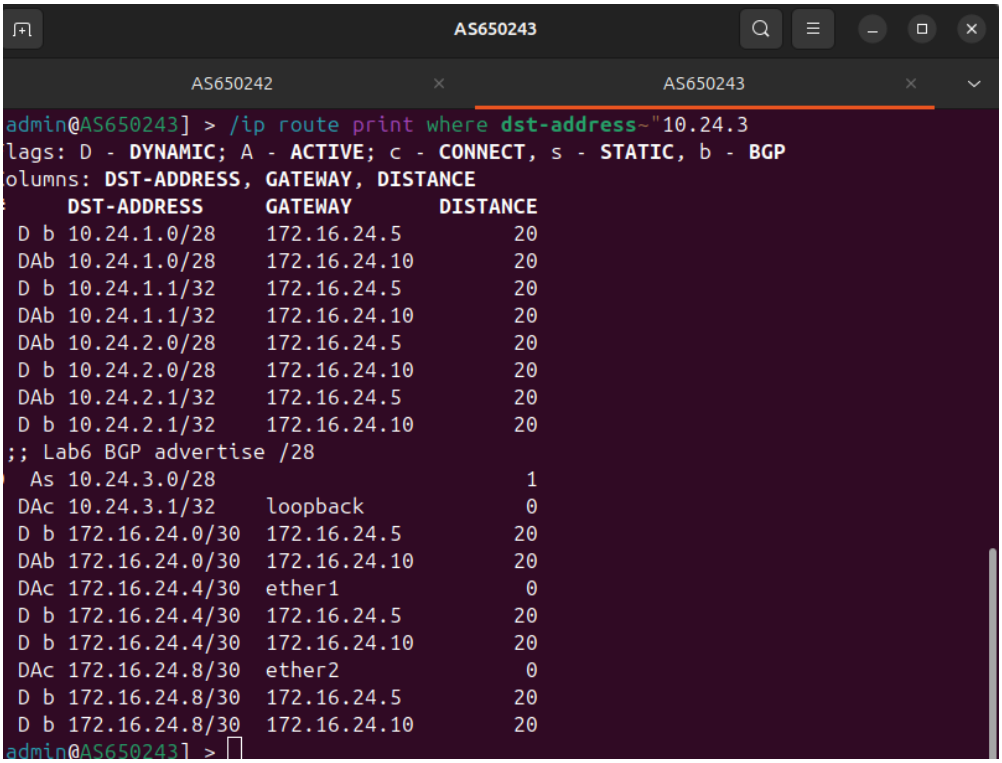
**dst-address~"10.24.1"**

```
[admin@AS650241] > /ip route print where dst-address~"10.24.1
Flags: D - DYNAMIC; I - INACTIVE, A - ACTIVE; c - CONNECT, s - STATIC, b - BGP>
Columns: DST-ADDRESS, GATEWAY, DISTANCE
#      DST-ADDRESS      GATEWAY         DISTANCE
0  IsH 0.0.0.0/0        192.168.8.1            1
1  IsH 0.0.0.0/0        192.168.8.181          1
2  IsH 0.0.0.0/0        192.168.8.1            1
;;; Lab6 BGP advertise /28
3  As  10.24.1.0/28                            1
  DAc  10.24.1.1/32     loopback               0
  DAb  10.24.2.0/28     172.16.24.2           20
  D b  10.24.2.0/28     172.16.24.9           20
  DAb  10.24.2.1/32     172.16.24.2           20
  D b  10.24.2.1/32     172.16.24.9           20
  D b  10.24.3.0/28     172.16.24.2           20
  DAb  10.24.3.0/28     172.16.24.9           20
  D b  10.24.3.1/32     172.16.24.2           20
  DAb  10.24.3.1/32     172.16.24.9           20
  DIcH 10.92.243.0/24   ether3                 0
  DAc  172.16.24.0/30   ether1                 0
  D b  172.16.24.0/30   172.16.24.2           20
  DAb  172.16.24.4/30   172.16.24.2           20
  D b  172.16.24.4/30   172.16.24.9           20
  DAc  172.16.24.8/30   ether2                 0
-- [Q quit|D dump|right|down]
```

**dst-address~"10.24.2"**



**dst-address~"10.24.3"**




**6. Ensure that BGP is advertising networks that exist in the routing table**

**Answer:**

In the lab topology, there were supposed to be two types of networks for each AS:

- Connected route: 10.24.x.1/32 (Loopback IP)
- Static route: 10.24.x.0/28 (Blackhole aggregate)

Three commands are used on each AS to verify that this requirement is met correctly:

- /ip route print detail where bgp
- /ip route print where bgp
- /routing bgp connection print detail

**AS650241**

**First, the following command is run on AS650241:**

```
[admin@AS650241] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.2
         immediate-gw=172.16.24.2%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.9
         immediate-gw=172.16.24.9%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.2.1/32 routing-table=main gateway=172.16.24.2
         immediate-gw=172.16.24.2%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.1/32 routing-table=main gateway=172.16.24.9
         immediate-gw=172.16.24.9%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.9
         immediate-gw=172.16.24.9%ether2 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.2
         immediate-gw=172.16.24.2%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.3.1/32 routing-table=main gateway=172.16.24.9
         immediate-gw=172.16.24.9%ether2 distance=20 scope=40 target-scope=10
```

The output here shows that AS650241 has learned and is advertising the following networks via BGP:

- *10.24.2.0/28* and *10.24.2.1/32*
- *10.24.3.0/28* and *10.24.3.1/32*

The gateways for these routes are *172.16.24.2* and *172.16.24.9*, indicating that the routes are being received from two EBGP neighbors (ECMP / redundancy).

**Then run for a brief view:**

```
[admin@AS650241] > /ip route print where bgp
Flags: D - DYNAMIC; A - ACTIVE; b - BGP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY      DISTANCE
DAb 10.24.2.0/28    172.16.24.2        20
D b 10.24.2.0/28    172.16.24.9        20
DAb 10.24.2.1/32    172.16.24.2        20
D b 10.24.2.1/32    172.16.24.9        20
DAb 10.24.3.0/28    172.16.24.9        20
D b 10.24.3.0/28    172.16.24.2        20
DAb 10.24.3.1/32    172.16.24.9        20
D b 10.24.3.1/32    172.16.24.2        20
D b 172.16.24.0/30  172.16.24.2        20
DAb 172.16.24.4/30  172.16.24.2        20
D b 172.16.24.4/30  172.16.24.9        20
D b 172.16.24.8/30  172.16.24.9        20
```

Here we see the same destination addresses, which match exactly with the previous detail output. That is, the networks that BGP is advertising and are active in the routing table.

**Then to verify BGP peering, we run:**

```
[admin@AS650241] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650242"
     remote.address=172.16.24.2 .as=650242
     local.address=172.16.24.1 .role=ebgp
     routing-table=main router-id=10.24.1.1 templates=TPL-AS650241 as=650241
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650243"
     remote.address=172.16.24.9 .as=650243
     local.address=172.16.24.10 .role=ebgp
     routing-table=main router-id=10.24.1.1 templates=TPL-AS650241 as=650241
     output.redistribute=connected,static .network=yes
```

Here we can see that:

• EBGP session with AS650242 (*172.16.24.2*)

• EBGP session with AS650243 (*172.16.24.9*)

• *output.redistribute=connected,static* and *network=yes* are enabled

This means that AS650241 is advertising both its connected and static (blackhole) routes in BGP.

**AS650242**

**Running same way on AS650242:**

```
[admin@AS650242] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.1/32 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10
```

AS65024 This output shows that the following networks are present as BGP routes:

- 10.24.1.0/28 and 10.24.1.1/32
- 10.24.3.0/28 and 10.24.3.1/32

These come from different gateways (*172.16.24.1* and *172.16.24.6*), which means that the routes are being learned from EBGP neighbors on both sides.

**Similarly, in 2, we first run:**

**In short, to verify it is run:**

```
[admin@AS650241] > /ip route print where bgp
Flags: D - DYNAMIC; A - ACTIVE; b - BGP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS      GATEWAY      DISTANCE
DAb 10.24.2.0/28     172.16.24.2        20
D b 10.24.2.0/28     172.16.24.9        20
DAb 10.24.2.1/32     172.16.24.2        20
D b 10.24.2.1/32     172.16.24.9        20
DAb 10.24.3.0/28     172.16.24.9        20
D b 10.24.3.0/28     172.16.24.2        20
DAb 10.24.3.1/32     172.16.24.9        20
D b 10.24.3.1/32     172.16.24.2        20
D b 172.16.24.0/30   172.16.24.2        20
DAb 172.16.24.4/30   172.16.24.2        20
D b 172.16.24.4/30   172.16.24.9        20
D b 172.16.24.8/30   172.16.24.9        20
```

The same network list is seen, meaning that the networks advertised in BGP are valid in the routing table.

**Then the BGP session is verified:**

```
[admin@AS650242] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650241"
     remote.address=172.16.24.1 .as=650241
     local.address=172.16.24.2 .role=ebgp
     routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650243"
     remote.address=172.16.24.6 .as=650243
     local.address=172.16.24.5 .role=ebgp
     routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
     output.redistribute=connected,static .network=yes
```

Here we can see that both AS650241 and AS650243 have EBGP sessions established and connected and static route redistribution enabled. This means that AS650242 has also advertised only existing routes.

**AS650243**

**Then it is verified on AS650243. First:**

```
[admin@AS650243] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10
```

Here we see that the following networks are present via BGP:

- *10.24.1.0/28* and *10.24.1.1/32*
- *10.24.2.0/28* and *10.24.2.1/32*

The gateways are *172.16.24.10* and *172.16.24.5*, meaning that the route is coming from two EBGP neighbors.

**Then** **briefly:**

```
[admin@AS650243] > /ip route print where bgp
Flags: D - DYNAMIC; A - ACTIVE; b - BGP
Columns: DST-ADDRESS, GATEWAY, DISTANCE
    DST-ADDRESS     GATEWAY       DISTANCE
DAb 10.24.1.0/28    172.16.24.10        20
D b 10.24.1.0/28    172.16.24.5         20
DAb 10.24.1.1/32    172.16.24.10        20
D b 10.24.1.1/32    172.16.24.5         20
D b 10.24.2.0/28    172.16.24.10        20
DAb 10.24.2.0/28    172.16.24.5         20
D b 10.24.2.1/32    172.16.24.10        20
DAb 10.24.2.1/32    172.16.24.5         20
DAb 172.16.24.0/30  172.16.24.10        20
D b 172.16.24.0/30  172.16.24.5         20
D b 172.16.24.4/30  172.16.24.10        20
D b 172.16.24.4/30  172.16.24.5         20
D b 172.16.24.8/30  172.16.24.10        20
D b 172.16.24.8/30  172.16.24.5         20
```

This command also shows the same network, which confirms that the advertised network and the routing table match each other.

**Finally, the BGP connection is verified:**

```
[admin@AS650243] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650241"
     remote.address=172.16.24.10 .as=650241
     local.address=172.16.24.9 .role=ebgp
     routing-table=main router-id=10.24.3.1 templates=TPL-AS650243 as=650243
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650242"
     remote.address=172.16.24.5 .as=650242
     local.address=172.16.24.6 .role=ebgp
     routing-table=main router-id=10.24.3.1 templates=TPL-AS650243 as=650243
     output.redistribute=connected,static .network=yes
```

Here, this can be seen that EBGP peering is established and connected with AS650241 and AS650242 and static route redistribution is enabled.

**Ping Test (End-to-End Verification)**

A ping test is performed to verify connectivity.

**AS650241**

```
[admin@AS650241] > /ping 10.24.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.1.1                                  56  64 45us
    1 10.24.1.1                                  56  64 120us
    2 10.24.1.1                                  56  64 123us
    3 10.24.1.1                                  56  64 173us
    4 10.24.1.1                                  56  64 169us
    sent=5 received=5 packet-loss=0% min-rtt=45us avg-rtt=126us max-rtt=173us
```

```
[admin@AS650241] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 1ms123us
    1 10.24.2.1                                  56  64 2ms165us
    2 10.24.2.1                                  56  64 2ms39us
    3 10.24.2.1                                  56  64 2ms248us
    4 10.24.2.1                                  56  64 2ms87us
    5 10.24.2.1                                  56  64 2ms195us
    6 10.24.2.1                                  56  64 742us
    7 10.24.2.1                                  56  64 2ms52us
    sent=8 received=8 packet-loss=0% min-rtt=742us avg-rtt=1ms831us
   max-rtt=2ms248us
```

```
[admin@AS650241] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 1ms123us
    1 10.24.2.1                                  56  64 2ms165us
    2 10.24.2.1                                  56  64 2ms39us
    3 10.24.2.1                                  56  64 2ms248us
    4 10.24.2.1                                  56  64 2ms87us
    5 10.24.2.1                                  56  64 2ms195us
    6 10.24.2.1                                  56  64 742us
    7 10.24.2.1                                  56  64 2ms52us
    sent=8 received=8 packet-loss=0% min-rtt=742us avg-rtt=1ms831us
   max-rtt=2ms248us
```

## AS650242

```
[admin@AS650242] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 39us
    1 10.24.2.1                                  56  64 51us
    2 10.24.2.1                                  56  64 163us
    3 10.24.2.1                                  56  64 125us
    4 10.24.2.1                                  56  64 126us
    5 10.24.2.1                                  56  64 133us
    sent=6 received=6 packet-loss=0% min-rtt=39us avg-rtt=106us max-rtt=163us
```

```
[admin@AS650242] > /ping 10.24.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.1.1                                  56  64 706us
    1 10.24.1.1                                  56  64 2ms108us
    2 10.24.1.1                                  56  64 1ms974us
    sent=3 received=3 packet-loss=0% min-rtt=706us avg-rtt=1ms596us
   max-rtt=2ms108us
```

```
[admin@AS650242] > /ping 10.24.3.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.3.1                                  56  64 1ms710us
    1 10.24.3.1                                  56  64 2ms11us
    2 10.24.3.1                                  56  64 2ms91us
    3 10.24.3.1                                  56  64 2ms79us
    4 10.24.3.1                                  56  64 2ms148us
    5 10.24.3.1                                  56  64 2ms55us
    sent=6 received=6 packet-loss=0% min-rtt=1ms710us avg-rtt=2ms15us
   max-rtt=2ms148us
```

## AS650243

```
[admin@AS650243] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 1ms449us
    1 10.24.2.1                                  56  64 1ms907us
    2 10.24.2.1                                  56  64 1ms953us
    3 10.24.2.1                                  56  64 2ms133us
    4 10.24.2.1                                  56  64 2ms116us
    5 10.24.2.1                                  56  64 1ms998us
    6 10.24.2.1                                  56  64 2ms38us
    sent=7 received=7 packet-loss=0% min-rtt=1ms449us avg-rtt=1ms942us
   max-rtt=2ms133us
```

```
[admin@AS650243] > /ping 10.24.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.1.1                                  56  64 913us
    1 10.24.1.1                                  56  64 2ms102us
    2 10.24.1.1                                  56  64 2ms94us
    3 10.24.1.1                                  56  64 2ms174us
    sent=4 received=4 packet-loss=0% min-rtt=913us avg-rtt=1ms820us
   max-rtt=2ms174us
```

```
[admin@AS650243] > /ping 10.24.3.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.3.1                                  56  64 73us
    1 10.24.3.1                                  56  64 130us
    2 10.24.3.1                                  56  64 132us
    3 10.24.3.1                                  56  64 128us
    4 10.24.3.1                                  56  64 194us
    5 10.24.3.1                                  56  64 127us
    6 10.24.3.1                                  56  64 44us
    7 10.24.3.1                                  56  64 126us
    8 10.24.3.1                                  56  64 126us
    9 10.24.3.1                                  56  64 124us
    sent=10 received=10 packet-loss=0% min-rtt=44us avg-rtt=120us
   max-rtt=194us
```

All ping replies were successfully received from all three ASs, which proves that there is no mismatch between the BGP advertisement and the routing table.

**Part C: Verification and Testin**

**1. Check BGP neighbor establishment on all routers**

**Answer:**

This topology is an EBGP-based lab topology, so each router must have an EBGP session up with its neighboring ASs. If a BGP neighbor is not established on a router, subsequent routing advertisements and connectivity will not work properly.

This command is used specifically in MikroTik RouterOS for this verification. Running this command clearly shows the remote AS, local address, role (ebgp), routing table, router-id, and redistribute configuration of each BGP neighbor.

```
[admin@AS650241] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650242"
     remote.address=172.16.24.2 .as=650242
     local.address=172.16.24.1 .role=ebgp
     routing-table=main router-id=10.24.1.1 templates=TPL-AS650241 as=650241
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650243"
     remote.address=172.16.24.9 .as=650243
     local.address=172.16.24.10 .role=ebgp
     routing-table=main router-id=10.24.1.1 templates=TPL-AS650241 as=650241
     output.redistribute=connected,static .network=yes
```

After running the above command on the AS650241 router, it is seen that there are two EBGP neighbors configured. The first neighbor is AS650242, whose remote address is *172.16.24.2* and local address is *172.16.24.1*. The second neighbor is AS650243, whose remote address is *172.16.24.9* and local address is *172.16.24.10.* In both cases, the role is set to ebgp and *output.redistribute=connected,static* and *network=yes* enable are visible. There is no inactive or disabled flag, which clearly indicates that both EBGP neighbor sessions with AS650241 are successfully established.

**Then, the same command is run on the AS650242 router:**

```
[admin@AS650242] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650241"
     remote.address=172.16.24.1 .as=650241
     local.address=172.16.24.2 .role=ebgp
     routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650243"
     remote.address=172.16.24.6 .as=650243
     local.address=172.16.24.5 .role=ebgp
     routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
     output.redistribute=connected,static .network=yes
```

Here we see AS650242:

- AS650241 (remote address *172.16.24.1*, local address *172.16.24.2*)
- AS650243 (remote address *172.16.24.6*, local address *172.16.24.5*)

Both of these EBGP neighbors are active and there is no inactive or disabled status. The Router-ID is set correctly and the options to redistribute connected and static routes are enabled. This indicates that BGP neighbor establishment for AS650242 is correct.

**Then the same verification is performed on the AS650243 router:**

```
[admin@AS650242] > /routing bgp connection print detail
Flags: D - dynamic, X - disabled, I - inactive
 0   name="EBGP-AS650241"
     remote.address=172.16.24.1 .as=650241
     local.address=172.16.24.2 .role=ebgp
     routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
     output.redistribute=connected,static .network=yes

 1   name="EBGP-AS650243"
     remote.address=172.16.24.6 .as=650243
     local.address=172.16.24.5 .role=ebgp
     routing-table=main router-id=10.24.2.1 templates=TPL-AS650242 as=650242
     output.redistribute=connected,static .network=yes
```

This router shows:

- EBGP session with AS650241 (remote *172.16.24.10*)
- EBGP session with AS650242 (remote *172.16.24.5*)

Both neighbors are active in the ebgp role and use routing table main. There is no error, inactive or disabled indication. As a result, the BGP neighbor relationship of AS650243 is also successfully established.

**2. Examine the BGP table to verify route learning**

**Answer:**

The BGP table should contain routes that are not from the local router, but rather learned from other ASs. This is most clearly seen on MikroTik when the route flag is b (bgp).

The same command was used on all routers in the lab to perform this verification:

The following command only shows routes that have been learned via the BGP protocol.

```
[admin@AS650242] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.6
         immediate-gw=172.16.24.6%ether2 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.0/28 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.3.1/32 routing-table=main gateway=172.16.24.1
         immediate-gw=172.16.24.1%ether1 distance=20 scope=40 target-scope=10
```

After running the command on router AS650241, the routing table shows the following networks:

- *10.24.2.0/28* and *10.24.2.1/32*
- *10.24.3.0/28* and *10.24.3.1/32*

All these routes have the b flag, which clearly indicates that they are BGP-learned routes.

The gateways are *172.16.24.2* (AS650242) and *172.16.24.9* (AS650243), meaning that AS650241 is learning routes from its two neighboring ASs.

This proves that AS650241 is successfully learning BGP routes.

**The same command is then run on router AS650242:**

```
[admin@AS650243] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10
```

Here in the routing table we see the following networks:

- *10.24.1.0/28* and *10.24.1.1/32*
- *10.24.3.0/28* and *10.24.3.1/32*

These routes clearly have the b flag next to them, which confirms that they were learned via BGP. The gateways are *172.16.24.1* (AS650241) and *172.16.24.6* (AS650243). This means that AS650242 is receiving routes correctly from both of its neighboring ASs and BGP route learning is fully effective.

**The command is then run on the AS650243 router:**

```
[admin@AS650243] > /ip route print detail where bgp
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10
```

The routing table of this router shows:

- *10.24.1.0/28* and *10.24.1.1/32*
- *10.24.2.0/28* and *10.24.2.1/32*

Each of these has the b flag, which directly proves that these routes are BGP-learned.

The gateways are *172.16.24.10* (AS650241) and *172.16.24.5* (AS650242). This means that AS650243 is also successfully learning routes from its two neighboring ASs.

### 3. Check the global routing table for BGP-learned routes

**Answer:**

The purpose of this part was to verify that the routes learned from BGP are not only installed in the BGP table, but also in the Global Routing Table (main routing table) of the router. Because in reality, traffic forwarding is always done according to the global routing table. If a route is learned from BGP but not installed in the global table, then traffic will not actually go through that route.

**The following command was used to perform this verification:**

```
[admin@AS650243] > /ip route print detail where dst-address~"10.24."
Flags: D - dynamic; X - disabled, I - inactive, A - active;
c - connect, s - static, r - rip, b - bgp, o - ospf, i - is-is, d - dhcp, v - >
H - hw-offloaded; + - ecmp
   DAb   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   DAb   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   D b   dst-address=10.24.1.1/32 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10

   D b   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.10
         immediate-gw=172.16.24.10%ether2 distance=20 scope=40
         target-scope=10

   DAb   dst-address=10.24.2.0/28 routing-table=main gateway=172.16.24.5
         immediate-gw=172.16.24.5%ether1 distance=20 scope=40 target-scope=10
```

This command displays all routes with the 10.24.x.x prefix from the global routing table, including connected, static, and BGP-learned routes.

After running the command on router AS650243, it is seen that the following routes are present in the global routing table:

- *10.24.1.0/28*
- *10.24.1.1/32*
- *10.24.2.0/28*

Two types of flags are seen for these routes:

- *DAb* = means the route is Dynamic + Active + BGP-learned
- *Db* = means Dynamic + BGP-learned

This means that these networks have been learned from BGP and installed in the global routing table.

*172.16.24.10* and *172.16.24.5* are seen as gateways, which indicates that the routes come from AS650241 and AS650242.

At the same time, both connected (/32) and BGP (/28) routes are visible together in the routing table, which proves that the global routing table is populated correctly.

**4. Test connectivity between all loopback interfaces**

**Answer:**

The loopback IP is usually used as the most stable interface of the router (for router-ID, BGP source, management, etc.), so if all the loopback IPs are reachable to each other, then it can be assumed that BGP routing, route advertisement and global routing table are all working properly. To verify this connectivity, the ping command is used from each router to the loopback IPs of the other two routers.

**AS650241**

Here, *10.24.2.1* is the loopback IP of AS650242 and *10.24.3.1* is the loopback IP of AS650243. Both pings received successful replies, which proves that there is connectivity from AS650241 to AS650242 and AS650243 in both directions.

```
[admin@AS650241] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 1ms123us
    1 10.24.2.1                                  56  64 2ms165us
    2 10.24.2.1                                  56  64 2ms39us
    3 10.24.2.1                                  56  64 2ms248us
    4 10.24.2.1                                  56  64 2ms87us
    5 10.24.2.1                                  56  64 2ms195us
    6 10.24.2.1                                  56  64 742us
    7 10.24.2.1                                  56  64 2ms52us
    sent=8 received=8 packet-loss=0% min-rtt=742us avg-rtt=1ms831us
   max-rtt=2ms248us
```


```
[admin@AS650241] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 1ms123us
    1 10.24.2.1                                  56  64 2ms165us
    2 10.24.2.1                                  56  64 2ms39us
    3 10.24.2.1                                  56  64 2ms248us
    4 10.24.2.1                                  56  64 2ms87us
    5 10.24.2.1                                  56  64 2ms195us
    6 10.24.2.1                                  56  64 742us
    7 10.24.2.1                                  56  64 2ms52us
    sent=8 received=8 packet-loss=0% min-rtt=742us avg-rtt=1ms831us
   max-rtt=2ms248us
```


Here, *10.24.2.1* is the loopback IP of AS650242 and *10.24.3.1* is the loopback IP of AS650243. Both pings received successful replies, which proves that there is connectivity from AS650241 to AS650242 and AS650243 in both directions.

**AS650242**

**The following ping test is then run from the AS650242 router:**

```
[admin@AS650242] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 39us
    1 10.24.2.1                                  56  64 51us
    2 10.24.2.1                                  56  64 163us
    3 10.24.2.1                                  56  64 125us
    4 10.24.2.1                                  56  64 126us
    5 10.24.2.1                                  56  64 133us
    sent=6 received=6 packet-loss=0% min-rtt=39us avg-rtt=106us max-rtt=163us
```

```
[admin@AS650242] > /ping 10.24.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.1.1                                  56  64 706us
    1 10.24.1.1                                  56  64 2ms108us
    2 10.24.1.1                                  56  64 1ms974us
    sent=3 received=3 packet-loss=0% min-rtt=706us avg-rtt=1ms596us
   max-rtt=2ms108us
```

```
[admin@AS650242] > /ping 10.24.3.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.3.1                                  56  64 1ms710us
    1 10.24.3.1                                  56  64 2ms11us
    2 10.24.3.1                                  56  64 2ms91us
    3 10.24.3.1                                  56  64 2ms79us
    4 10.24.3.1                                  56  64 2ms148us
    5 10.24.3.1                                  56  64 2ms55us
    sent=6 received=6 packet-loss=0% min-rtt=1ms710us avg-rtt=2ms15us
   max-rtt=2ms148us
```

Here, the pings to both the loopback IPs *10.24.1.1* (AS650241) and *10.24.3.1* (AS650243) returned successful replies. This means that the AS650242 router is able to reach the loopbacks of both neighboring routers end-to-end.

**AS650243**

**The following ping test is performed from the AS650243 router:**

```
[admin@AS650243] > /ping 10.24.2.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.2.1                                  56  64 1ms449us
    1 10.24.2.1                                  56  64 1ms907us
    2 10.24.2.1                                  56  64 1ms953us
    3 10.24.2.1                                  56  64 2ms133us
    4 10.24.2.1                                  56  64 2ms116us
    5 10.24.2.1                                  56  64 1ms998us
    6 10.24.2.1                                  56  64 2ms38us
    sent=7 received=7 packet-loss=0% min-rtt=1ms449us avg-rtt=1ms942us
   max-rtt=2ms133us
```

```
[admin@AS650243] > /ping 10.24.1.1
  SEQ HOST                                     SIZE TTL TIME       STATUS
    0 10.24.1.1                                  56  64 913us
    1 10.24.1.1                                  56  64 2ms102us
    2 10.24.1.1                                  56  64 2ms94us
    3 10.24.1.1                                  56  64 2ms174us
    sent=4 received=4 packet-loss=0% min-rtt=913us avg-rtt=1ms820us
   max-rtt=2ms174us
```

Here too, the ping is successful. This means that the loopback interface of AS650243 to AS650241 and AS650242 - two routers-reachable.

**5. Perform traceroute between different ASes to verify the path taken**

**Answer:**

The purpose of this part was to verify that the packet is taking a path from one AS to the loopback IP of another AS, that is, whether the AS-to-AS routing path is working properly. Ping only shows connectivity, but traceroute shows the exact hop the packet is taking to reach the destination. The topology uses three ASs (AS650241, AS650242, AS650243) connected to each other via EBGP. Traceroute is run from each router to the loopback IP of the other router.

**AS650242 - AS650241 (Traceroute Test)**

**First, traceroute is run from router AS650242 to the loopback IP (*10.24.1.1*) of AS650241:**

```
[admin@AS650242] > /tool traceroute 10.24.1.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
#  ADDRESS    LOSS  SENT  LAST  AVG  BEST  WORST  STD-DEV
1  10.24.1.1  0%      10  2ms   1.9  0.6   2.1    0.4
```

The traceroute output shows that the first hop *10.24.1.1* is found and there is no packet loss (0% loss). This means that the routing path from AS650242 to AS650241 is working directly and correctly, and the route learned by BGP is correctly installed in the global routing table.

**AS650241 - AS650242 (Traceroute Test)**

**A traceroute is then run from the AS650241 router to the loopback IP (*10.24.2.1*) of AS650242:**

```
[admin@AS650241] > /tool traceroute 10.24.2.1
Columns: ADDRESS, LOSS, SENT, LAST, AVG, BEST, WORST, STD-DEV
#  ADDRESS    LOSS  SENT  LAST   AVG  BEST  WORST  STD-DEV
1  10.24.2.1  0%       4  2.1ms  1.8  0.8   2.1    0.6
```

Here again, the traceroute output shows that the destination IP is reachable in one hop and there is no loss. This proves that the BGP path from AS650241 to AS650242 is resolving correctly and the routing is symmetric.

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